

DANE COUNTY REGIONAL, WI, USA

WMO: 726410

Lat: 43.141N

Lon: 89.345W

Elev: 264

StdP: 98.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14837

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.5	-18.8	-26.7	0.3	-20.6	-23.9	0.4	-18.1	10.2	-5.3	9.0	-4.6	2.4	300	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	31.8	23.3	30.2	22.5	28.8	21.7	24.9	30.0	23.8	28.5	22.9	27.4	4.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.6	28.3	22.3	17.5	27.0	21.3	16.5	25.8	77.0	30.0	72.7	28.7	68.8	27.5	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.4	-25.3	34.2	3.5	1.9	-27.8	35.6	-29.9	36.7	-31.9	37.7	-34.4	39.1		
(5)			-25.6	26.5	3.5	1.3	-28.0	27.4	-30.1	28.2	-32.0	28.9	-34.5	29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-7.0	-4.9	1.4	8.2	14.5	20.2	22.3	21.1	17.0	10.2	2.9	-3.6
	DBStd	11.43	6.74	6.55	6.39	5.14	4.69	3.77	3.26	3.09	4.42	4.95	5.44	6.14
	HDD10.0	2029	527	418	276	92	11	0	0	0	3	59	220	423
	HDD18.3	3918	786	651	525	305	137	24	4	9	76	256	463	681
	CDD10.0	1517	0	1	11	38	152	305	381	343	214	66	7	1
	CDD18.3	363	0	0	1	2	20	78	127	94	37	4	0	0
	CDH23.3	3272	0	0	6	27	198	737	1192	751	325	37	0	0
	CDH26.7	934	0	0	0	3	46	223	388	189	80	5	0	0
Wind	WSAvg	3.3	3.5	3.6	3.8	4.2	3.5	3.0	2.7	2.5	2.8	3.2	3.5	3.4
Precipitation	PrecAvg	852	32	32	56	85	91	110	99	101	85	63	54	44
	PrecMax	1288	73	84	157	181	275	278	237	264	234	143	190	104
	PrecMin	537	4	2	7	34	16	8	28	22	3	2	3	6
	PrecStd	156	18	20	34	35	52	69	42	56	57	36	34	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.4	13.5	23.1	26.7	30.5	33.4	35.2	32.4	31.5	27.3	19.8	13.9
		MCWB	6.2	9.5	16.0	16.5	20.8	23.4	24.9	24.1	22.1	19.8	13.9	12.2
	2%	DB	5.8	8.8	17.9	22.9	27.8	31.0	32.0	30.5	28.9	23.8	16.2	9.8
		MCWB	3.4	5.7	12.4	14.9	19.1	22.6	24.1	23.0	21.4	17.4	11.9	7.1
	5%	DB	3.4	6.0	14.1	20.1	25.6	29.2	30.3	28.7	26.9	21.2	13.7	6.6
		MCWB	1.9	3.7	10.2	12.6	18.1	21.4	22.9	22.2	19.9	16.1	10.1	4.7
	10%	DB	1.8	3.4	10.7	17.3	23.2	27.5	28.8	27.3	25.0	18.6	11.3	4.2
		MCWB	0.7	1.6	6.8	11.0	16.6	20.6	22.2	21.2	18.7	14.4	8.1	2.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	9.9	17.0	18.0	22.6	25.1	27.0	25.7	23.9	21.2	14.9	12.7
		MCDB	8.0	12.5	22.7	23.6	28.9	30.8	32.0	30.8	28.7	25.5	18.2	13.9
	2%	WB	3.8	6.3	13.8	15.7	20.7	23.8	25.4	24.4	22.4	19.1	12.8	7.7
		MCDB	5.5	8.4	17.1	21.4	25.7	29.3	30.7	28.5	26.8	22.2	15.6	8.7
	5%	WB	2.0	3.7	10.2	13.9	19.2	22.7	24.1	23.3	21.2	16.9	10.8	4.9
		MCDB	3.1	5.9	13.8	19.0	23.7	28.0	28.9	27.2	25.2	20.1	13.1	6.4
	10%	WB	0.7	1.7	6.9	11.6	17.7	21.5	23.1	22.2	20.1	14.8	8.2	2.5
		MCDB	1.8	3.3	10.4	16.3	22.0	26.0	27.4	25.9	24.0	18.3	10.8	4.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.6	9.8	11.1	11.3	10.9	10.7	10.4	11.6	10.3	8.5	7.4
		MCDBR	8.9	10.4	13.7	15.3	13.7	12.6	11.7	11.3	13.3	13.4	12.1	9.3
		MCWBR	7.5	7.9	8.9	8.9	7.1	6.2	5.4	5.5	6.5	8.0	8.8	7.7
	5% WB	MCDBR	8.3	10.1	12.4	13.4	11.8	11.1	10.3	9.7	10.9	11.2	11.2	8.6
		MCWBR	7.2	7.9	8.8	8.7	6.9	6.2	5.7	5.5	6.1	8.1	9.0	7.7
Clear-Sky Solar Irradiance	taub	0.274	0.312	0.325	0.368	0.395	0.412	0.415	0.400	0.369	0.329	0.293	0.272	
	taud	2.390	2.278	2.399	2.314	2.290	2.279	2.270	2.306	2.382	2.492	2.524	2.441	
	Ebn at Noon	869	880	918	897	879	861	853	855	858	857	846	839	
	Edh at Noon	77	103	104	123	129	131	131	122	105	83	68	67	
All-Sky Solar Radiation	RadAvg	1.70	2.73	3.79	4.55	5.38	6.03	6.08	5.28	4.31	2.77	1.80	1.31	
	RadStd	0.22	0.29	0.42	0.40	0.39	0.44	0.31	0.34	0.35	0.33	0.16	0.11	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (57 neighbors)		(m)								
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page